6.2 Introduction to Conics: Parabolas

Name _____ Period ____ Date _____

Pre-Calculus 6.2 · Student sheet

I can…

- write the standard equation of a parabola.
- find the vertex, focus, and directrix of a parabola.
- use parabolas to solve real-life problems.

Vocabulary · fill in the missing word

Conic section — the intersection of a plane and a _____ cone

Parabola — the set of points _____ from a fixed line (the directrix) and a fixed point (the focus)

Axis of a parabola — the line through the focus _____ to the directrix

Part 1 · Fold a parabola

Mark the focus about 1 inch above the bottom edge. Fold the edge onto the focus 20–30 times.

Distance from focus to edge	Shape the creases make	Where is the vertex?

Part 2 · Notes and examples (fill in each note, then work the example)

Standard form (vertex (h, k)): vertical axis $(x - h)^2 = 4p(y - k)$; horizontal axis $(y - k)^2 =$ _____.

Direction: $p > 0$ opens up or right; $p < 0$ opens _____.

Example 1

Write the standard equation of the parabola with vertex (0, 0) and focus (2, 0).

Focus and directrix: the focus is $|p|$ units from the vertex inside the curve; the directrix is $|p|$ units the _____ way.

Example 2

Find the vertex, focus, and directrix of $y = -\frac{1}{2}x^2 - x + \frac{1}{2}$.

Example 3

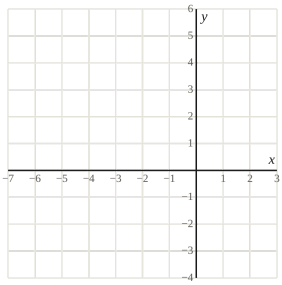
Write the standard equation with vertex (2, 1) and focus (2, 4).

Example 4

A headlight reflector is a parabola 6 inches wide and 1 inch deep. How far from the vertex should the bulb (focus) go?

Example 5

Sketch $(y - 1)^2 = -8(x + 2)$. Give the vertex, focus, and directrix.



Part 3 · Partner practice

Partner 1

Find the vertex, focus, and directrix of $x^2 = -12y$.

Partner 2

Write the standard equation with vertex $(-1, 3)$ and directrix $x = -3$.